

Prediction of diabetes using logistic regression and ensemble techniques

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ABSTRACT

Background: Logistic regression is a classification model in machine learning, extensively used in clinical analysis. It uses probabilistic estimations which helps in understanding the relationship between the dependent variable and one or more independent variables. Diabetes, being one of the most common diseases around the world, when detected early, may prevent the progression of the disease and avoid other complications. In this work, we design a prediction model, that predicts whether a patient has diabetes, based on certain diagnostic measurements included in the dataset, and explore various techniques to boost the performance and accuracy.

Methods: Logistic Regression is the main algorithm used in this paper and the analysis is carried out using Python IDE. The experiment mainly uses two datasets – one is the PIMA Indians Diabetes dataset, which is originally from the National Institute of Diabetes and Digestive and Kidney Diseases, and the other dataset is from Vanderbilt, which is based on a study of rural African Americans in Virginia. Feature selection is carried out using two different methods. Ensemble methods are further used, that improve performance by producing better predictions compared to a single model.

Results: The accuracy and runtimes are captured for the original datasets and also for the ones obtained after using feature selection and ensemble techniques. A comparison is also shown in each case. The highest accuracy obtained was around 78% for Dataset 1, after employing the ensemble technique- Max Voting; and it was around 93% for Dataset 2, after using the ensemble techniques- Max Voting, and Stacking.

Conclusion: Logistic Regression has shown to be one of the efficient algorithms in building prediction models. This study also shows that apart from the choice of algorithms, there are other factors that could improve the accuracy and runtimes of the model, such as: data-preprocessing, removal of redundant and null values, normalization, cross-validation, feature selection, and usage of ensemble techniques.

1. Introduction

Diabetes, also known as Diabetes Mellitus, targets many people around the world. According to the International Diabetes Federation, approximately 463 million adults (20–79 years) were living with diabetes in 2019. They predicted that by 2045 this will rise to 700 million. Diabetes prevalence has been rising more rapidly in low- and middle-income countries than in high-income countries. Diabetes is a major cause of blindness, kidney failure, heart attacks, stroke and lower limb amputation [1]. It is also estimated that around 84.1 million Americans who are 18 years or older have prediabetes [2].

There are three types of Diabetes. Type-1 is known as Insulin-Dependent Diabetes Mellitus (IDDM). The reason behind this type of diabetes is the inability of a human's body to generate enough insulin. In this case, the patient is required to inject insulin. Type-2 is also known as Non-Insulin-Dependent Diabetes Mellitus (NIDDM). This type of

Diabetes is seen when body cells are not able to use insulin properly. Type-3 Gestational Diabetes increases the blood sugar level in pregnant woman [3]. This happens when diabetes is not detected in the early stages. Even though Diabetes is incurable, it can be managed by treatment and medication.

Many healthcare organizations are now using Machine Learning Techniques, such as Predictive Modeling in healthcare. Additionally, there are complex algorithms at play, identifying processes and patterns invisible to the human eye. This helps researchers discover new medicine and treatment plans. Predictive modeling uses data mining, machine learning, and statistics to identify patterns in data and recognize the chances of outcomes occurring.

This paper focuses on building a predictive model for diabetes to identify if a certain patient has diabetes and then various techniques are explored to improve accuracy. Logistic Regression will be used to develop the main model, and the first dataset used is the PIMA Indian

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Dataset [4]. In this dataset, all patients are females who are at least 21 years old. The paper explains the step by step process of the model, - from its design to its implementation. The second dataset used is from Vanderbilt [5], which is based on a study of rural African Americans in Virginia. It consists of 16 features. This dataset consists of both male and female patients.

The rest of the paper is organized as follows: Section 2 discusses the previous work and how this work differs from theirs. Sections 3 and 4 consist of the model design, methods used and results. Section 5 lists the challenges which are yet to be addressed and finally Section 6 concludes the overall experiment.

2. Literature review

Machine Learning Techniques are becoming more useful in the medical sector. Many researchers have used various Machine Learning and Deep Learning Techniques and Algorithms to predict diabetes.

Aishwarya and Vaidehi [3] used several machine learning algorithms such as Support Vector Machines, Random Forest Classifier, Decision Tree Classifier, Extra Tree Classifier, Ada Boost algorithm, Perceptron, Linear Discriminant Analysis algorithm, Logistic Regression, K-NN, Gaussian Naïve Bayes, Bagging algorithm and Gradient Boost Classifier. They used two different datasets- the PIMA Indian and another Diabetes dataset for testing the various models. Logistic Regression gave them an accuracy value of 96%. On the other hand, Tejas and Pramila [6] chose two algorithms- Logistic Regression and SVM to build a diabetes prediction model. The pre-processing of data was carried out to obtain better results. They found that SVM performed better with an accuracy of 79%.

Yuvaraj and Sripreetha [7] designed a diabetes prediction model using three different Machine Learning algorithms- Random Forest, Decision Tree, and the Naïve Bayes, in Hadoop based clusters. They employed pre-processing techniques on the dataset. The results showed that the highest accuracy rate of 94% was obtained with the Random Forest algorithm. Deepti and Dilip [8] used Decision Tree, SVM, and Naïve Bayes algorithms. Ten-fold cross validation was used to improve performance. The highest accuracy was obtained by the Naïve Bayes, with an accuracy of 76.30%. Both these papers used the Pima Indian Diabetes dataset.

Both, Olaniyi and Adnan [9], and Swapna et al. [10] made use of Deep Learning techniques for diabetes prediction. The former made use of a Multilayer Feed-Forward Neural Network. The back-propagation algorithm was used for training the model. They also used the PIMA Indian dataset and normalized it before pre-processing, to obtain numerical stability. They obtained 82% accuracy. The latter used a dataset called Electrocardiograms on two models using CNN and CNN-LSTM. The dataset consisted of 142,000 samples and eight attributes. They obtained an accuracy of 93.6% with the CNN model and an accuracy of 95.1% for the CNN-LSTM model with a five-fold cross validation for both.

All the above studies gave a comparative performance analysis among various machine learning algorithms. Some of them used data pre-processing and cross-validation techniques to improve accuracy, but all of them focused more on the comparison of performance between the various models rather than improving a single model. In this paper, I have concentrated on a single model and explored techniques which can not only improve accuracy but also improve execution speed, thus increasing the performance. This paper shows that in addition to algorithm selection, pre- and post- processing of data play a major role in the overall improvement of the model.

3. Model design

Two main datasets are used in this paper. The first one is the PIMA Indian Dataset, which consists of 768 patient's data who are all females of the age 21 or older. There are nine features in total. The second

dataset is from Vanderbilt, which is based on a study of rural African Americans in Virginia. It consists of 16 features. There are 390 data samples with both male and female patients. Here Dataset 1 refers to the PIMA Indian Dataset and Dataset 2 refers to the dataset from Vanderbilt. The main algorithm used in the prediction model is Logistic Regression, although few other machine learning techniques such as Decision Tree, Support Vector Machines, K- nearest Neighbors and Naïve Bayes are used in the ensemble methods to test the improvement in the original performance. We have designed a flowchart of the prediction model, which is shown in Fig. 1. It shows the flow of how the implementation will be carried out.

Various methods are explored to improve the performance and execution time. Firstly, it starts with two feature selection methods - creating new features and then selecting the best; and the other method is Univariate Feature Selection [11]. The second technique is using ensemble methods. In this project two ensemble methods, namely- Max Voting/Majority Voting and Stacking are used. All the results are analyzed using the IDE PyCharm with Python 3.6 version on the Windows 10 platform.

4. Methods and results

4.1. Dataset selection and feature evaluation

The first steps involve selecting the dataset for the model and evaluating its features. For this paper, the first dataset chosen is the PIMA Indian dataset. There are a total of nine features/variables, among which eight are predictor variables and 1 is the target variable. The features are as follows:

- Pregnancies: Number of times the patient was pregnant.
- Glucose: Plasma glucose concentration over two hours in an oral glucose tolerance test.
- BloodPressure: Diastolic blood pressure (mm Hg).
- SkinThickness: Triceps skin fold thickness (mm).
- Insulin: Two-Hour serum insulin (μ U/ml).
- BMI: Body mass index ($\text{weight in kg}/(\text{height in m})^2$).
- DiabetesPedigreeFunction/DPF: A function that scores the likelihood of diabetes based on family history.
- Age: In years.
- Outcome: Class variable (0 if non-diabetic, 1 if diabetic). This is the target variable.

The second dataset used is the Vanderbilt dataset. It consists of 16 features, out of which one is the target variable, i.e. Diabetes:

- Patient number: Identifies patients by number
- Cholesterol: Total cholesterol
- Glucose: Fasting blood sugar
- HDL: HDL or good cholesterol
- Chol/HDL: Ratio of total cholesterol to good cholesterol. Desirable result is < 5
- Age: Age of the patient
- Gender: 162 males, 228 females
- Height: In inches
- Weight: In pounds (lbs)
- BMI: $703 \times \text{weight (lbs)} / [\text{height (inches)}]^2$
- Systolic BP: The upper number of blood pressure
- Diastolic BP: The lower number of blood pressure
- Waist: Measured in inches
- Hip: Measured in inches
- Waist/hip: Ratio is possibly a stronger risk factor for heart disease than BMI
- Diabetes: Yes (60), No (330)

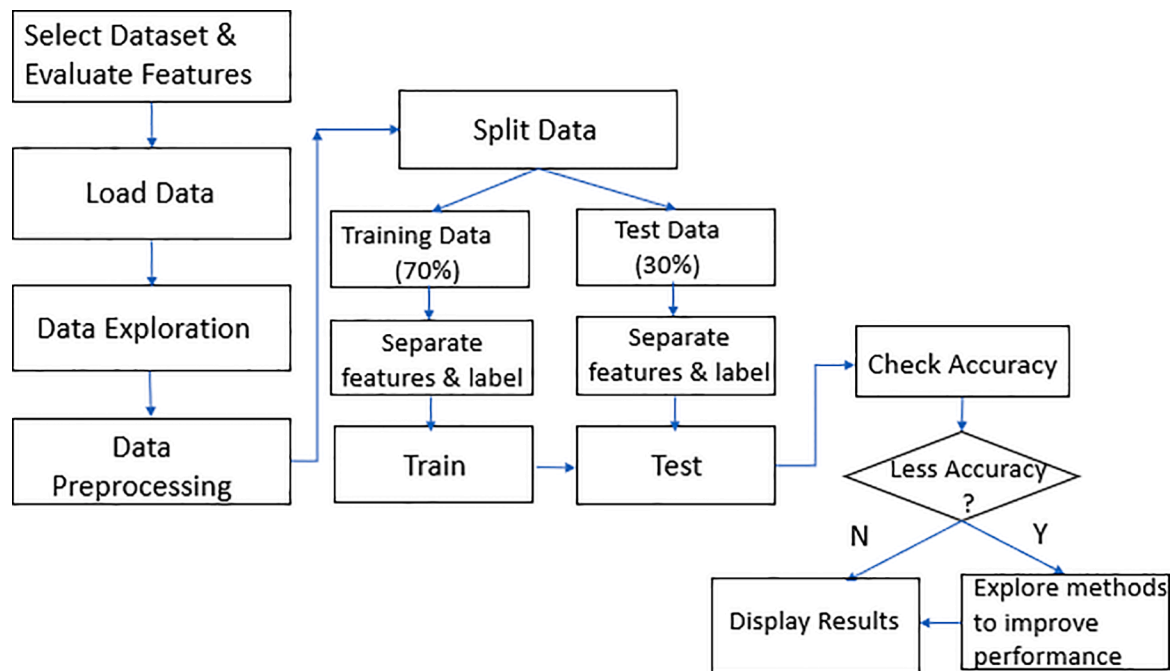


Fig. 1. Flowchart of the Diabetes Prediction Model.

4.2. Loading data

The data, which is in the CSV format, is loaded to a variable. There are 768 data points in Dataset 1 and 390 data points in Dataset 2.

4.3. Data exploration

Data exploration involves getting insights about the data and finding the correlation between the features. Dataset 1 consists of 268 patients

with diabetes and the remaining 500 without diabetes, whereas the Dataset 2 has 60 diabetic patients and 330 who are not diabetic. The heatmap displayed in Fig. 2, shows the correlation between the features of Dataset 1. The lighter colors represent more correlation and the darker colors represent less correlation. The Fig. 3 shows a bar plot displaying count of patients with and without Diabetes in Dataset 1.

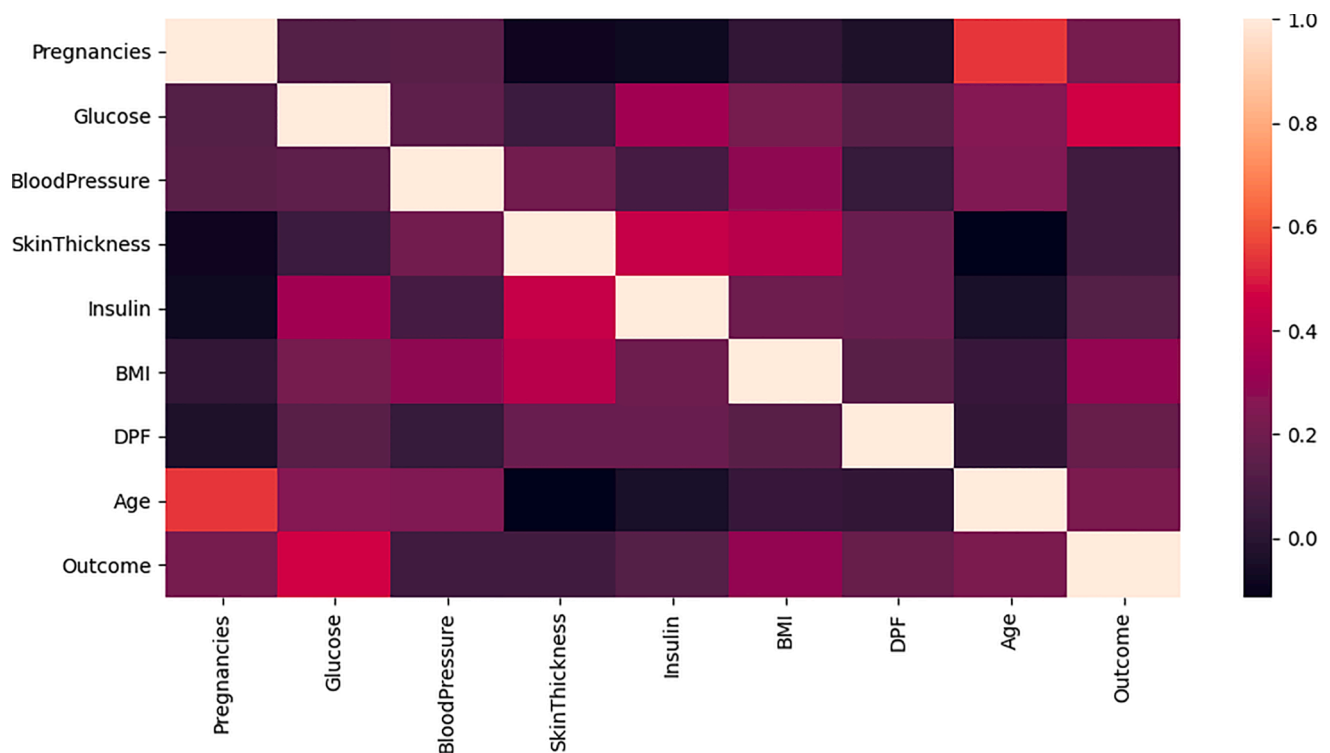


Fig. 2. Correlation values and heat map, showing the correlation between the features for Dataset 1.

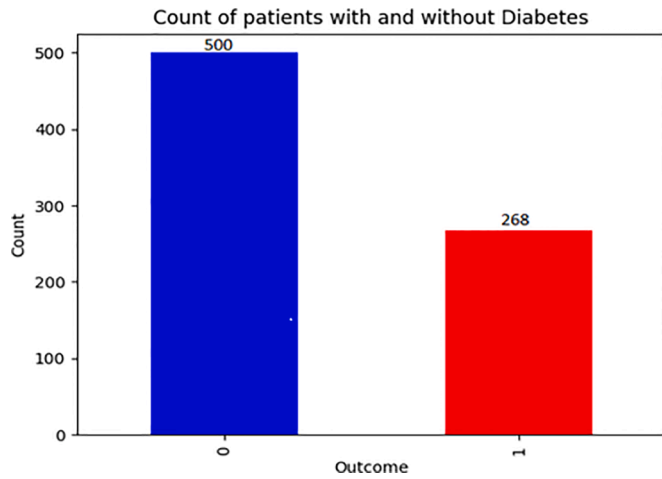


Fig. 3. xxx.

4.4. Data pre-processing

The missing values, null values and values equal to zero for the predictor variables need to be identified in the dataset. The predictor variables/features cannot have a zero value except for certain features, like for example the 'Pregnancies' feature in Dataset 1. These values need to be replaced with the mean values of the column. This is an important step to increase the accuracy of prediction, as faulty values increase the chances of incorrect predictions.

4.5. Training and testing

The data is split into training and testing sets. The common split ratios are 80:20 and 70:30. In this project, the data is split in the ratio of 70:30, i.e. 70% for training and 30% for testing. A Logistic Regression algorithm is used to make the predictions and check for the accuracy. The execution time is also calculated.

For predictions, there are four important terms: - True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN). TP and TN represent the cases when the actual outcome and the result are the same, whereas FP and FN are the cases when the opposite results are obtained. A classification report is generated which includes Precision, Recall, F1 score, and Support. The Precision metric shows what percent of predictions are correct. Recall describes what percent of positives are correctly identified. The F1 score is the percent of positive predictions that are correct. Support is the count of actual occurrences of the class in the specified dataset. Table 1 shows the classification report for both the datasets:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1 Score} = 2 * (\text{Recall} * \text{Precision}) / (\text{Recall} + \text{Precision})$$

Table 1
Classification report.

For Dataset 1	Precision	Recall	F1 score	Support
0	0.77	0.85	0.81	151
1	0.66	0.53	0.58	80
Accuracy			0.74	231
Macro Avg.	0.71	0.69	0.70	231
Weighted Avg.	0.73	0.74	0.73	231
For Dataset 2	Precision	Recall	F1 score	Support
0	0.90	0.96	0.93	93
1	0.78	0.58	0.67	24
Accuracy			0.88	117
Macro Avg.	0.84	0.77	0.80	117
Weighted Avg.	0.87	0.88	0.87	117

4.6. Improving accuracy

Various methods are employed to improve the accuracy. The methods include: - creating new features; feature selection using Univariate feature selection; and ensemble methods, such as Max Voting and Stacking.

4.6.1. Creating new features

Five new features are created from the existing features and added to the dataset. These features are created based on a research of certain diagnostic measurements applicable to diabetic patients. The heatmap is plotted and based on the correlation with the output and with each other, eight features have been selected. With these eight features, the accuracy is again calculated. The accuracy seemed to increase. For this method, Dataset 1 is used.

The new features are labelled as NF1, NF2, NF3, NF4 and NF5. The first parameter NF1 is basically chosen on the basis that, usually, people above the age of 30 are less prone to this disease. Also, the blood sugar level below 140 mg/dL is normal [12]. NF2 is for individuals with BMI higher than 30 kg/m², who are at a higher risk of diabetes [13]. NF3 is chosen based on the study carried out by Chengjie Lv et al. that women with pregnancies greater or equal to four are at a greater risk of diabetes than the women with number of pregnancies three or less [14]. The normal diastolic blood pressure is less than or equal to 80 and this is the main factor considered for feature NF4. The last feature NF5 is a combination of a normal glucose value and a higher BMI value. These features are described as follows:

- NF1 - Age less than or equal to 30 and Glucose value less than or equal to 140
- NF2 - BMI is less than or equal to 30
- NF3 - Age less than or equal to 30 and Pregnancies is less than or equal to three
- NF4 - Glucose value less than or equal to 140 and Blood Pressure less than or equal to 80
- NF5 - Glucose value less than or equal to 140 and BMI less than or equal to 45

The correlation between all the old and new features are calculated. Fig. 4 shows the heat map with the correlation values between the features.

Depending on the correlation values, eight features are selected in such a way that they have less correlation among each other and more correlation with the output. The extra features are deleted and only the selected eight features are retained. Fig. 5 shows the heat map with only the retained values.

The training and testing are again carried out using these new features, and the accuracy and execution time are noted. Table 2 shows the classification report with the new features. Table 3 also shows the difference in execution times and accuracy before and after using this method. We can observe an increase in the accuracy and a major change in the runtime of the program, which decreased to a great extent. This highlights the importance of feature selection [15] in improving model performance.

4.6.2. Univariate feature selection

Univariate Feature selection [16] is a method that works by selecting the best features based on univariate statistical tests. Each feature is compared to the target variable and any statistically significant relationship between them is determined. It is also called analysis of variance. The feature selection is carried out using the Univariate feature selection, in particular, - using the chi-square test [17]. Chi-Square measures how the expected count and observed count deviate each other. The formula is given below:

$$(\chi_c)^2 = \sum (O_i - E_i)^2 / E_i$$

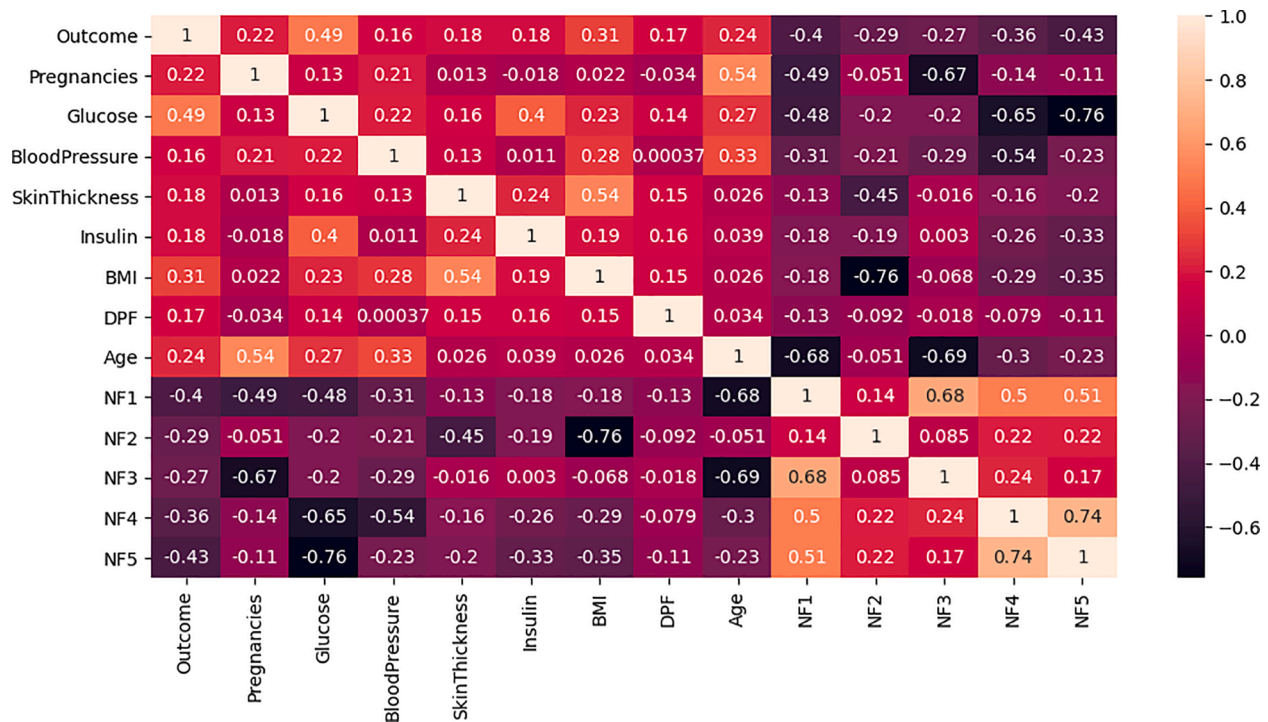


Fig. 4. Heat map showing the correlation between all the old and new features.

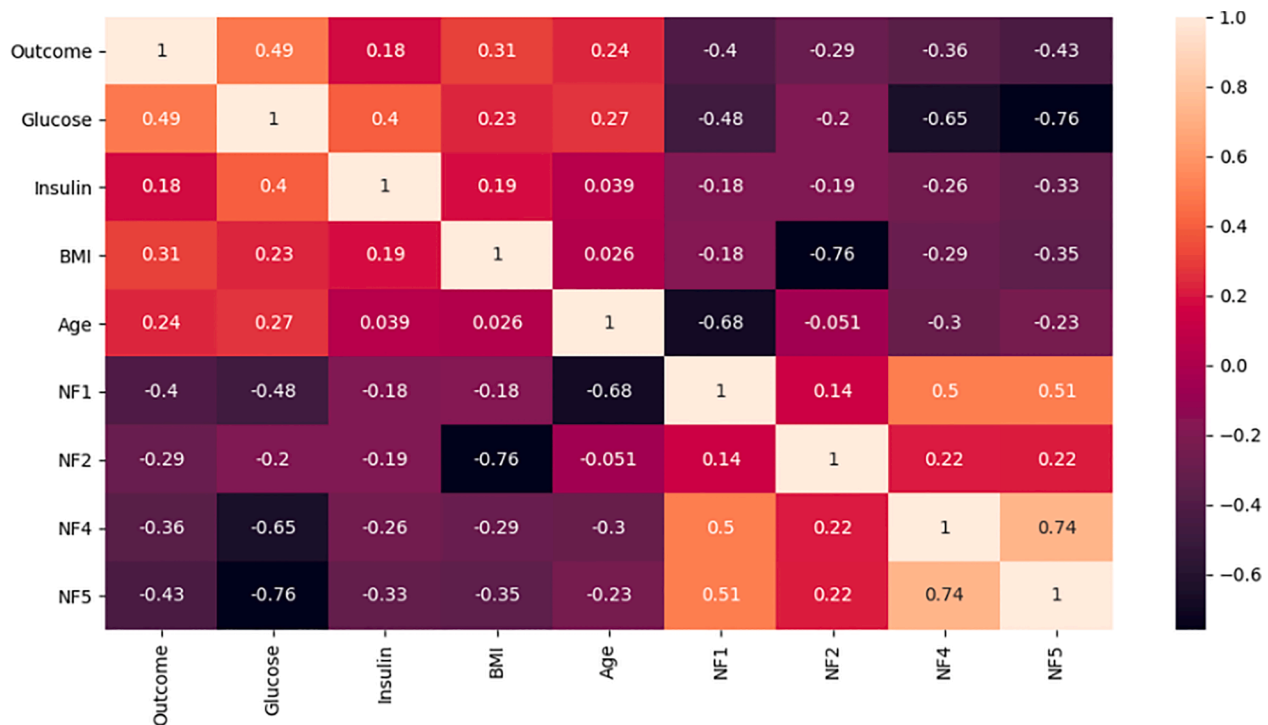


Fig. 5. Heat map with correlation values of retained features.

where:

c = degrees of freedom

O = observed value(s)

E = expected value(s)

The 'SelectKBest' and 'chi2' methods from the sklearn library are

used to achieve the univariate feature selection in Python. Eight highest scoring features are selected, and the test is carried out. The selected features are: – Cholesterol, Glucose, HDL Chol, Chol/HDL ratio, Age, Weight, Systolic BP and Waist. For this method, the second dataset from Vanderbilt is used. Training and testing using Logistic Regression are carried out with the selected features. The classification report after Univariate Feature selection can be observed in Table 4. In this method, there was a slight increase in the accuracy but significant decrease in the

Table 2

Classification report for dataset 1 after Feature Selection.

	Precision	Recall	F1 score	Support
0	0.81	0.82	0.81	151
1	0.65	0.62	0.64	80
Accuracy			0.75	231
Macro Avg	0.73	0.72	0.73	231
Weighted Avg	0.75	0.75	0.75	231

Table 3

Comparison before and after feature selection for Dataset 1.

	Accuracy	Execution Time
Before	0.7403	0.042
After	0.7532	0.008

Table 4

Classification report for dataset 2 after univariate feature selection.

	Precision	Recall	F1 score	Support
0	0.89	0.98	0.93	93
1	0.87	0.54	0.67	24
Accuracy			0.89	117
Macro Avg	0.88	0.76	0.80	117
Weighted Avg	0.89	0.89	0.88	117

runtime. This result can be verified from Table 5. Reduction of the runtime is essential especially when handling larger datasets or while handling time sensitive projects.

4.6.3. Ensemble methods

Ensemble methods/algorithms help to improve the machine learning results by combining several models. Hence the performance of the predictive models can be further improved by using such methods. Ensemble methods placed first in many prestigious machine learning competitions, such as the Netflix Competition, KDD 2009, and Kaggle [18]. In this project, two methods- Max Voting and Stacking are used, in an attempt to improve the performance, and the differences after using these methods are noted.

4.6.3.1. Max voting/majority voting. This is one of the simplest methods in ensemble techniques. It combines predictions from various machine learning algorithms. The predictions by each model are known as a 'vote'. The predictions which are obtained from the majority votes of the models are used as the final prediction [19]. In general, the formula for Max Voting [20] for classifiers $c_1, c_2, \dots, c_N$ is given as follows:

$$P(X) = \operatorname{argmax}_i \sum_{j=1}^N w_j p_{ij} \text{ where:}$$

$P(X)$ is the final prediction

w_j is the weight of the j th classification

p_{ij} is the probability estimate from the j th classification rule for i th class

We have used the 'VotingClassifier' from the sklearn library to carry out the Max Voting. This method is applied after feature selection is carried out in both datasets, in the hope of increasing the accuracy. Additionally, a ten-fold cross-validation is applied to add an additional boost to the performance.

Table 5

Comparison before and after feature selection for dataset 2.

	Accuracy	Execution Time
Before	0.8803	0.036
After	0.8889	0.005

In this experiment, we make use of five different machine learning algorithms: – Logistic Regression, Decision Tree, Support Vector Machines, K- nearest Neighbors and Naïve Bayes. The accuracy of each individual model, as well as the ensemble models for both datasets, are shown in Table 6. The max voting ensemble technique increased the accuracy of the models in both the cases.

4.6.3.2. Stacking. Stacking is a technique that uses predictions from multiple models to build a new model. The models used to build the new model are known as base models. The base models are learnt parallelly and combined by a meta-model or a final model to give the final prediction.

In this experiment, Decision Tree, Naïve Bayes and k-NN models are used as the base models and the Logistic Regression model is used as the final model for prediction. Cross- validation is essential with Stacking. This is because there is a limitation with this method, that is, all the base models contribute equally to the intermediate result, no matter how well a single model may perform. In this experiment, there is no improvement in accuracy without cross-validation. Once a three-fold cross-validation is carried out for the models used in stacking, there is an improvement in accuracy for both the datasets. Table 7 shows the accuracy values of the base models and the final model after k-fold cross-validation. Table 8, given below, shows the comparison of accuracy before and after Stacking.

4.7. Results summary

The predictive model was first designed with only the Logistic Regression algorithm after pre-processing the null values and eliminating the missing values. The feature selection techniques were later employed to help improve the accuracy and execution time. For Dataset 1, new features were created from existing ones and a correlation heat map was plotted to select the best eight features. For Dataset 2, Univariate feature selection using chi-square test, was employed, choosing eight top-scoring features. Ensemble methods were further used to try and boost performance. Max/Majority Voting and Stacking methods were tested on both the datasets. The former method proved to be a best method among all, by showing a significant improvement in performance. The latter performed well after cross- validation was incorporated.

Table 8, given below, shows the complete summary of the various techniques used in this experiment to build and improve the model, with their accuracy values. Figs. 6 and 7 show the accuracy bar plots of these techniques. LR represents Logistic Regression in both the figures.

5. Discussion

Prediction of diabetics using AI techniques has been addressed in the literature, but for the first time we adopt the regression method to this problem to improve the accuracy. In contrast to prior work [21,22], where the focus was on a single model, here several models were used using ensemble techniques to improve the performance. It was discovered that the performance does not just depend on the algorithm used but on other factors such as data-preprocessing, feature selection and the use of different models using ensemble techniques such as max/majority voting and stacking. The improvement in both, accuracy and execution time, was highlighted.

There are several challenges that come with using AI techniques. The technologies are still not extensively used in clinical practice. There is a requirement of larger datasets, so that more accurate and reliable models can be obtained. This can increase the complexity of the model. In addition, a single model may not be applicable to the overall population. The major challenges in disease-risk prediction modeling with the machine learning methods include the lack of reproducibility and external validation [21].

Table 6

Accuracy for individual and ensemble models.

	Logistic Regression	Decision Tree	Support Vector Machine	k-Nearest Neighbor	Naïve Bayes	Ensemble Model
Dataset 1	0.7532	0.7229	0.7489	0.6797	0.7403	0.7783
Dataset 2	0.8889	0.8803	0.8803	0.8803	0.8889	0.9341

Table 7

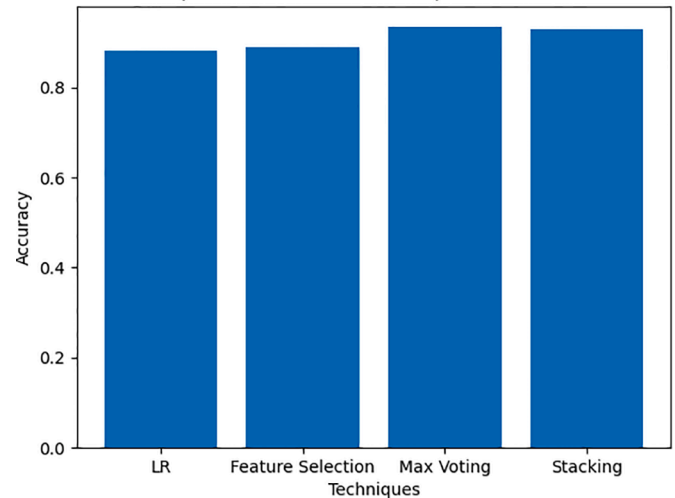
Comparison of Accuracy with and without Stacking.

	Base Models	Accuracy after k-fold	Final Model Accuracy of Logistic Regression after k-fold
Dataset 1	Decision Tree	0.724	0.7635
	Naïve Bayes	0.748	
	k-NN	0.756	
	Decision Tree	0.923	
Dataset 2	Naïve Bayes	0.930	0.9304
	Bayes		
	k-NN	0.923	

Table 8

Comparison of Accuracy with and without Stacking Summary of all the techniques.

	Without Stacking	With Stacking
Dataset 1	0.7532	0.7635
Dataset 2	0.8889	0.9304
	Accuracy of Dataset 1	Accuracy of Dataset 2
Only with Logistic Regression	0.7403	0.8803
With Feature Selection	0.7532	0.8889
With Ensemble Technique – Max Voting	0.7783	0.9341
With Ensemble Technique – Stacking	0.7635	0.9304

Comparison of various techniques for Dataset 2**Fig. 7.** Bar plot depicting accuracies of various techniques for Dataset 2.

There are also challenges in finding the correct balance between the accuracy and execution speeds. With more emphasis given on improving accuracy, the complexity of the model may increase, and improvement of execution speeds may take a backseat. Finding the correct balance is essential, as both these factors affect the performance of the machine learning model.

6. Conclusion

Logistic Regression has been shown to be one of the efficient algorithms in building prediction models. The accuracy of the model does not only depend on the algorithm chosen, but also on other factors. Data-preprocessing is one such factor. Removing redundant and null values is essential in improving performance. The normalization of values, when the features differ by large scale, also plays an important role. In this paper, we have seen that feature selection plays a significant role in increasing the accuracy and reducing the runtime. Combining various algorithms, as observed in the ensemble techniques, serves as a factor in improving the performance of the model. Cross-validation also holds an essential role to boost accuracy.

Declaration of Competing interests

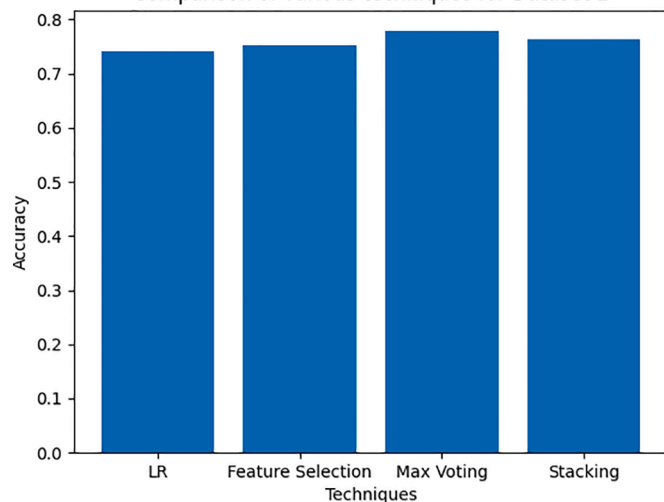
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Comparison of various techniques for Dataset 1**Fig. 6.** Bar plot depicting accuracies of various techniques for Dataset 1.

Another major challenge is the risk of patients' sensitive information being leaked. Wearable technology constantly records data on the behavior of diabetic users, and also encourages them to input their sensitive health data for it to provide more accurate, tailor-made suggestions. This would inherently expose users to the risk of having personal data leaked or being used for other purposes without their consent [22].

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